Seth M. Woodbury

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Professional Summary

Highly motivated bioengineering doctoral student at the University of Washington from a background in biochemistry, physics, and chemistry with a creative passion for integrating different scientific disciplines to develop innovative solutions. Extensive undergraduate research and development experience with inventing biomaterials and drug-delivery systems for tissue engineering. Pursuing doctoral research in computational enzyme design with a focus in *de novo* metalloenzyme design for applications in sustainable chemistry, green manufacturing, environmental remediation, and advanced therapeutics. Pioneering a future where scientists can custom-design highly active, ideal biocatalysts for any physics-allowed target reaction.

Education

University of Washington, Seattle, WA **June 2023 – Present**

Doctor of Philosophy (Ph.D.) Pre-Candidate:
Bioengineering — Certificate in Data-Science
GPA: 3.85/4.00

- Ongoing Thesis Topic: Computational Design of Metalloenzymes
- Advisor: David Baker

University of Michigan, Ann Arbor, MI **Aug. 2019 – Apr. 2023**

Bachelor of Science (B.S.) with Highest Honors, Triple Major:
Chemical Science — Interdisciplinary Physics — Biochemistry
GPA: 3.850/4.000

- Undergraduate Honors Thesis: *Multifunctional Biomaterial Technologies with Sustained Drug Delivery for Diverse Periodontal Regeneration Applications*
- Advisors: W. Benton Swanson, Yuji Mishina (P.I.)

Research Funding

Large Project Grants

Defense Advanced Research Projects Agency **2025 – 2026**

Project: Computationally Designed Nerve Agent Catalytic Sponges and Hydrolases
Award: $1,990,000

Personal Funding & Fellowships

National Science Foundation Graduate Research Fellowship Program **2023 – 2028**

Provides funding for my Ph.D. research in bioengineering at the University of Washington.

University of Michigan LSA Honors Program, Honors Grants **2021 – 2023**

Funded my Chemical Science Honors Thesis in the Mishina-Swanson lab.

University of Michigan UROP Scholars, Travel Grant **2022**

Covered travel expenses to attend the Society For Biomaterials Annual Meeting in Baltimore, MD.

University of Michigan UROP Scholars, Travel Grant **2022**

Supported registration and attendance at the AADR/IADR hybrid meeting in Atlanta, GA.

University of Michigan UROP, Research Grant **2021 – 2022**

Funded various research projects in the Mishina-Swanson lab.

Intellectual Property

1. Nanofibrous Tissue Engineering Matrices with Improved Clinical Handling Properties for Periodontal and Craniofacial Regeneration and Methods of Making the Same. Inventors: William Benton Swanson, **Seth Woodbury**, Yuji Mishina. Patent application filed US 18/141,138 (Feb. 22, 2024 — US20240058103A1)

Peer-Reviewed Publications

1. Kim, D.‡; **Woodbury, S. M.‡**; Ahern, W.‡; Tischer, D.; Kang, A.; Joyce, E.; Bera, A. K.; Hanikel, N.; Salike, S.; Krishna, R.; Yim, J.; Pellock, S. J.; Lauko, A.; Kalvet, I.; Hilvert, D.; Baker, D. Computational Design of Metallohydrolases. *Nature* **2025**, 1–8. doi:10.1038/s41586-025-09746-w.

2. Ahern, W.; Yim, J.; Tischer, D.; Salike, S.; **Woodbury, S. M.**; Kim, D.; Kalvet, I.; Kipnis, Y.; Coventry, B.; Altae-Tran, H. R.; Bauer, M. S.; Barzilay, R.; Jaakkola, T. S.; Krishna, R.; Baker, D. Atom-Level Enzyme Active Site Scaffolding Using RFdiffusion2. *Nat. Methods* **2025**, 1–10. doi:10.1038/s41592-025-02975-x.

3. Chen, A.; Wu, K.; Choi, H.; Venkatesh, P.; Pellock, S. J.; Hanikel, N.; Coventry, B.; Kim, D.; **Woodbury, S. M.**; Ji, P.; Honda, S.; Li, X.; Gerben, S.; Chang, L.; Yan, X.; Hyman, A. A.; Hilvert, D.; Baker, D. Computational Design of Metalloproteases. *bioRxiv* **2025**, p 2025.11.20.689622. doi:10.1101/2025.11.20.689622 — under peer review at *Nature*.

4. Swanson, W. B.‡; **Woodbury, S. M.‡**; Dal-Fabbro, R.; Douglas, L.; Albright, J.; Eberle, M.; Niemann, D.; Xu, J.; Bottino, M. C.; Mishina, Y. Synthetic Periodontal Guided Tissue Regeneration Membrane with Self-Assembling Biphasic Structure and Temperature-Sensitive Shape Maintenance. *Adv. Healthc. Mater.* **2025**, *14*(3), 2402137. doi:10.1002/adhm.202402137

5. **Woodbury, S. M.**; Swanson, W. B.; Douglas, L.; Niemann, D.; Mishina, Y. Temperature-Responsive PCL-PLLA Nanofibrous Tissue Engineering Scaffolds with Memorized Porous Microstructure Recovery. *Front. Dent. Med.* **2023**, *4*. doi:10.3389/fdmed.2023.1240397

6. **Woodbury, S. M.**; Swanson, W. B.; Mishina, Y. Mechanobiology-Informed Biomaterial and Tissue Engineering Strategies for Influencing Skeletal Stem and Progenitor Cell Fate. *Front. Physiol.* **2023**, *14*. doi:10.3389/fphys.2023.1220555

7. Swanson, W. B.; Mahmoud, A. H.; **Woodbury, S. M.**; Bottino, M. C. Methacrylated Gelatin as an On-Demand Injectable Vehicle for Drug Delivery in Dentistry. In *Oral Biology: Molecular Techniques and Applications*; Seymour, G. J.; Cullinan, M. P.; Heng, N. C. K.; Cooper, P. R., Eds.; Springer US: New York, NY, 2023; pp 493–503. doi:10.1007/978-1-0716-2780-8_30

8. Swanson, W. B.; Durdan, M.; Eberle, M.; **Woodbury, S. M.**; Mauser, A.; Gregory, J.; Zhang, B.; Niemann, D.; Herremans, J.; Ma, P. X.; Lahann, J.; Weivoda, M.; Mishina, Y.; Greineder, C. F. A Library of Rhodamine6G-Based pH-Sensitive Fluorescent Probes with Versatile in Vivo and in Vitro Applications. *RSC Chem. Biol.* **2022**, *3*(6), 748–764. doi:10.1039/D2CB00030J

9. Swanson, W. B.; Omi, M.; **Woodbury, S. M.**; Douglas, L. M.; Eberle, M.; Ma, P. X.; Hatch, N. E.; Mishina, Y. Scaffold Pore Curvature Influences MSC Fate through Differential Cellular Organization and YAP/TAZ Activity. *Int. J. Mol. Sci.* **2022**, *23*(9), 4499. doi:10.3390/ijms23094499

Honors & Awards

Georgia Tech President's Fellowship Recipient (Declined) **2023**
Awarded to conduct a Ph.D. in the Biomedical Engineering Program at Georgia Institute of Technology.

Dental/Craniofacial Biomaterials Special Interest Group (SIG) Poster Prize Award **2022**
Awarded by the Society For Biomaterials.

Organic Chemistry II Study Group Facilitator of the Year **2022**
Awarded by the Science Learning Center, University of Michigan.

Bioartography Showcase Selection **2022**
Selected for showcase at School of Dentistry Research Day, University of Michigan.

Blue Ribbon Award for Outstanding Research Presentation **2021**
Presented by the Undergraduate Research Opportunities Program, University of Michigan.

James B. Angell Scholar Award **2021**
Recognized for two or more consecutive semesters of all A grades at the University of Michigan.

Alumni First Year Achievement Award **2020**
Awarded for exceptional performance in first-year chemistry laboratory and lecture work, University of Michigan.

William J. Branstrom Freshman Prize **2020**
Awarded to students in the top 5% of the freshman class at the end of the year, University of Michigan.

Tailhook Education Foundation (TEF) Scholarship Recipient **2020 – 2022**

Michigan Education Association Scholarship Recipient **2020 – 2022**

University Honors **2019 – 2022**
Recognized for academic excellence, University of Michigan.

Senior Scholar Athlete CAAC Award **2019**
Presented by the State of Michigan.

Advanced Placement Scholar with Distinction **2018 – 2019**
Awarded by the CollegeBoard®.

Employment & Professional Development

Honors Resident Advisor & DEI Event Planner **Aug. 2022 – Apr. 2023**
University of Michigan LSA Honors College
- Fostering a positive environment for 85 diverse undergraduates.
- Planning events to promote inclusivity and community within the Honors Program.
- Mentoring students to maximize their academic experience.

Organic Chemistry, Physics, & Biology Study Group Facilitator **Apr. 2020 – Apr. 2023**
University of Michigan Science Learning Center
- Leading weekly study sessions, preparing learning materials, and mentoring students.
- Creating a welcoming environment that encourages student engagement and learning.

Division I Athlete STEM Tutor **Aug. 2021 – Aug. 2022**
University of Michigan Academic Success Program
- Provided one-on-one tutoring for 6-8 student-athletes in STEM subjects.
- Adhered to NCAA rules and maintained confidentiality with high-profile individuals.

Undergraduate Research Opportunity Program (UROP) Scholar **Aug. 2020 – Apr. 2022**
University of Michigan
- Conducted extensive research in the Mishina-Swanson lab.
- Participated in workshops and training to enhance scientific communication skills.

Calculus Tutor & Guided Review Instructor **Feb. 2020 – May 2021**
Campus Tutors LLC
- Led calculus exam review sessions for 60+ students.
- Developed materials that blended lecture and active participation.

President & Co-Founder, Astrobiology Club **Dec. 2019 – Aug. 2020**
University of Michigan
- Founded an organization focused on biology and astrophysics.
- Drafted a NASA proposal for spacecraft sterility research.

Conference Presentations

1. **Woodbury, S.M.**, Swanson, W.B., Mishina, Y. PLLA-Based Biomaterials with PCL and PLGA Semi-IPNs Exhibiting Novel Thermally-Induced Nanofiber Formation, Thermosensitive Shape-Memory, and Controlled Degrees of Phase Separation for Strategies in Tissue Engineering. University of Michigan URAN|UM Symposium, Aug. 11, 2022. Ann Arbor, MI. (Poster)

2. **Woodbury, S.M.**, Swanson, W.B., Douglas, L., Mishina, Y. Nanofibrous Tissue Engineering Matrices from Poly (L-Lactide) with Improved Clinical Handling Properties for Periodontal and Craniofacial Regeneration.
   - Society For Biomaterials, Apr. 27-30, 2022. Baltimore, MD. (Poster)
   - University of Michigan Undergraduate Research Symposium, Apr. 8, 2022. Ann Arbor, MI. (Poster)
   - University of Michigan School of Dentistry Research Day, Feb. 17, 2022. Ann Arbor, MI. (Poster)

3. Swanson W.B., Omi, M., **Woodbury, S.M.**, Nam, H.K., Ma, P.X., Hatch, N.E., Mishina, Y. Towards Regeneration of the Cranial Suture: Scaffold Design Regulates Mesenchymal Stem Cell Fate through YAP/TAZ Activation. Society For Biomaterials, Apr. 27-30, 2022. Baltimore, MD. (Poster)

4. **Woodbury, S.M.**, Swanson, W.B., Douglas, L., Mishina, Y. Nanofibrous Tissue Engineering Matrices with Improved Handling Properties and a Sustained Drug-Delivery System for Periodontal and Craniofacial Regeneration. University of Michigan UROP Scholars Symposium, Apr. 20, 2022. Ann Arbor, MI. (Oral)

5. **Woodbury, S.M.**, Swanson, W.B., Eberle, M., Mishina, Y. Computational Approach to Determining the Properties of Rhodamine-Based pH-Sensitive Fluorophores used in Diverse Biomedical Applications.
   - University of Michigan CHEM — UNITY Symposium, Apr. 7, 2022. Ann Arbor, MI. (Virtual Oral)
   - University of Michigan 1st Year UROP Symposium, Apr. 22, 2021. Ann Arbor, MI. (Virtual Oral)

6. Swanson W.B., Omi, M., **Woodbury, S.M.**, Nam, H.K., Ma, P.X., Hatch, N.E., Mishina, Y. Towards Regeneration of the Cranial Suture: Pore Curvature Regulates Mesenchymal Stem Cell Fate in Macroporous Tissue Engineering Scaffolds. American Association of Dental, Oral, and Craniofacial Research, Mar. 23, 2022. Atlanta, GA. (Virtual Oral)

7. **Woodbury, S.M.**, Swanson, W.B., Mishina, Y. Nanofibrous, Thermoresponsive Tissue Engineering Scaffolds Demonstrating Macropore Recovery in Irregularly Shaped Defects. University of Michigan URAN|UM Symposium, Jul. 29, 2021. Ann Arbor, MI. (Virtual Oral)

8. **Woodbury, S.M.**, Gajda, M., Heng, J., Nayak, K., Golob, J., MacCready, K.G., Schmidt T. Supplement Consumption and pH in the Gut Microbiome. University of Michigan Biostatistics 201 Research Project Symposium, Dec. 9, 2019. Ann Arbor, MI. (Oral)

Technical Skills

Graduate Skills

Computational Chemistry
- Building models of molecular systems and conformer sampling
- Energy-based ground and transition state optimization of molecular systems

Computational Protein & Enzyme Design
- Biomolecular software engineering and design pipeline development
- Structure-based *de novo* protein and enzyme design with generative ML/AI methods
- Native enzyme redesign and sequence design
- Structure prediction

Experimental Molecular Biology & Biophysics
- DNA cloning and transformation
- Protein expression and chromatography purification (affinity, size-exclusion)

- Enzyme kinetics and mechanistic studies

Undergraduate Skills

Materials Chemistry & Fabrication
- Scaffold/construct design and fabrication
- Polymer synthesis, crosslinking, nanoparticle fabrication, and drug encapsulation
- Nanofibers and multiphasic membranes for tissue engineering

Materials Characterization
- Physical analysis (SEM, Contact Angle, XRD, DSC)
- Spectroscopy (FTIR, NMR, UV-Vis)
- Chromatography (HPLC, GC, SEC) & ICP-MS
- Drug/protein release kinetics

Biology & Cell Culture
- Histology, gel electrophoresis, and confocal microscopy
- Basic cell culture techniques and protein quantification

Quantitative & Computational Skills
- MATLAB, R, Spartan, and ChemDraw
- CAD (AutoCAD), GraphPad Prism, PyMOL, and ImageJ

Professional Memberships

Society For Biomaterials **2022 – 2023**

American Association of Dental Research **2022 – 2023**